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Review article

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Barren plateaus in variational quantum computing

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Abstract

Variational quantum computing offers a flexible computational approach with a broad range of applications. However, a key obstacle to realizing their potential is the barren plateau (BP) phenomenon. When a model exhibits a BP, its parameter optimization landscape becomes exponentially flat and featureless as the problem size increases. Importantly, all the moving pieces of an algorithm – choices of ansatz, initial state, observable, loss function and hardware noise – can lead to BPs if they are ill-suited. As BPs strongly impact on trainability, researchers have dedicated considerable effort to develop theoretical and heuristic methods to understand and mitigate their effects. As a result, the study of BPs has become a thriving area of research, influencing and exchanging ideas with other fields such as quantum optimal control, tensor networks and learning theory. This article provides a review of the current understanding of the BP phenomenon.

Sections

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Key points

- Variational quantum algorithms (VQAs) — this hybrid computational approach aims at training a quantum learning model (usually a parametrized quantum circuit) to solve a given task. The parameters in the model are trained by minimizing a loss function that encodes the degree to which the problem has been solved.

- Barren plateaus — a phenomenon in which the gradients of the loss landscape of VQAs get exponentially suppressed. Currently, this issue is understood as a form of curse of dimensionality arising from operating in an unstructured manner in an exponentially large Hilbert space.

- Trainability — in the context of VQAs, trainability refers to the ability to optimize parameters of a model and minimize the loss function. Barren plateaus are one of the main barriers to the trainability of VQAs.

Introduction

In recent years, variational quantum algorithms (VQAs) have gathered momentum as a promising computational approach[1–7]. At their core, most VQAs follow the same principle – convert a problem into an optimization task and use a classical device to train a parametrized quantum learning model. The popularity of this computational approach can be attributed in part to its flexibility, which allows access to applications in diverse areas in basic science and machine learning, but also gives rise to the hope that it may be more robust to the constraints of near-term quantum hardware.

In contrast to conventional quantum algorithms, which generally come with provable guarantees, VQAs share much of the heuristic nature of classical machine learning. But unlike classical machine learning, quantum computing lacks the hardware needed to test these algorithms at scale. Moreover, it quickly becomes clear to any practitioner that off-the-shelf variational schemes can be hard to train as the number of qubits increases. This has prompted efforts to analyse the scalability of these algorithms and to determine trainability barriers that could obstruct their practical use in problems of realistic sizes.

Although there are various sources for potential untrainability in VQAs, this Review focuses on the barren plateau (BP) phenomenon[8,9]. On a BP, the loss gradients, or more generally loss differences[10], vanish exponentially with the size of the system, which can cause problems with trainability. As information can only be extracted from a quantum computer through a finite set of measurements, the presence of a BP landscape implies that for the overwhelming majority of parameter choices an exponentially large number of measurement shots are needed to identify the loss minimizing direction.

By now, researchers have uncovered a number of causes of BPs, as well as several methods to mitigate or avoid them. The endeavour to understand this phenomenon has highlighted that BPs are a symptom of a deeper issue with variational quantum computing; namely, that the exponentially large Hilbert space dimension – the very same feature which was hoped to provide variational quantum computing with computational capabilities beyond those achievable by classical learning models – can cause problems if it is not handled properly. In essence, BPs ultimately arise from a 'curse of dimensionality'[11].

In this Review, we survey the body of literature on BPs that has emerged over the past 5 years. Our main goal is to provide guidelines on what can and what cannot help to avoid BPs that our community has identified. However, we also aim to show that striving to understand BPs can provide wider insights into the nature of the information processing capabilities of quantum computers, the resources required to make something 'quantum', the connection between the absence of BPs and classical simulability and the potential paths for alternative variational paradigms.

We start our Review with a brief review of variational quantum computing, before we revisit definitions for different types of BPs and identify their sources. We then survey architectures and methods that can and cannot mitigate BPs, or even attempt to prevent BPs altogether. In the next section, we present examples of the impact that the BP study has had beyond the field of variational quantum computing, and then move on to discuss other issues beyond BPs, such as local minima and the intriguing connection between absence of BPs and classical simulability. Next, we link the BP phenomenon with vanishing gradients in classical machine learning. We finish this Review with some closing remarks and an outlook into the future. Throughout the paper, we have added boxes that include important details and more technical discussions.

Variational quantum computing

Framework

The key idea behind variational quantum computing is to combine the power of quantum and classical computers. This hybrid approach leverages the large Hilbert space of the quantum device to process information and estimate quantities of interest. The results are then fed to a classical computer that analyses them and determines the next quantum experiment. Many such hybrid algorithms use an optimization or learning-based approach, in which a classical optimizer trains quantum experiments characterized by a sequence of parameters to solve a task. As variational computing reduces the burden on quantum hardware, it is well suited for near-term devices. However, their usage is not confined to the near term, and we expect variational schemes to flourish in the fault-tolerant era when noise is no longer an issue.

It has become standard to divide variational quantum computing schemes into two main categories – VQAs[1–3] and quantum machine-learning (QML) models[4–6]. VQAs can be considered as problem-driven, as the algorithms tackle a specific task such as preparing the ground state of some Hamiltonian; QML models, by contrast, are data-driven. QML has been envisioned for virtually all learning tasks such as supervised and unsupervised learning as well as generative modelling[6]. Throughout this Review, we will discuss VQAs and QML models together, as they mostly share a common structure.

Components of variational quantum computing

As shown in Fig. 1, in variational quantum computing, one first initializes the quantum system to some n-qubit initial state ρ, belonging to $\mathcal{B}(\mathcal{H}_0)$, the set of bounded linear operators on a 2^n-dimensional complex Hilbert space $\mathcal{H}_0$. In VQAs, ρ is usually an easy-to-prepare fiducial state, whereas in QML ρ is taken from a set (or distribution) of training states that encode either quantum or classical data. By quantum data, we mean that ρ is obtained from some quantum mechanical process of interest within a quantum device. By classical data, we mean data (such as images, financial information and events from a particle accelerator), and these data are intended to be processed in a quantum computer[1].

Next, one sends ρ through a parametrized quantum circuit (PQC), also known as a quantum neural network in the context of QML. A PQC can be expressed as a sequence of L parametrized unitaries

$$U(\boldsymbol{\theta}) = \prod_{i=1}^{L} U_i(\boldsymbol{\theta}_i), \qquad (1)$$

in which $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, ..., \boldsymbol{\theta}_L)$ is a set of trainable parameters. When hardware noise is present, or when the parameterized quantum circuit involves adding or removing qubits, the PQC is expressed more generally as a sequence of L parametrized channels:

$$\mathcal{U}_{\boldsymbol{\theta}} = \mathcal{U}_{\boldsymbol{\theta}_L} \circ \cdots \circ \mathcal{U}_{\boldsymbol{\theta}_1}, \quad (2)$$

in which each $\mathcal{U}_{\boldsymbol{\theta}_l} : \mathcal{B}(\mathcal{H}_{l-1}) \to \mathcal{B}(\mathcal{H}_l)$ is a completely positive and trace-preserving map between bounded operators. Finally, as indicated in Fig. 1, at the output of the PQC, one performs a (finite) number of measurements to estimate a quantity of interest, such as the expectation value of some set of observables or the probability of a given outcome.

In the simplest case, one has a single input state and is interested in estimating the expectation value of a single observable O, which leads to the loss function

$$\ell_{\boldsymbol{\theta}}(\rho, O) = \mathrm{Tr}[\rho(\boldsymbol{\theta})O], \quad (3)$$

with $\rho(\boldsymbol{\theta}) = \mathcal{U}_{\boldsymbol{\theta}}(\rho)$. Although we will discuss more general loss functions subsequently (some built by computing several quantities such as $\ell_{\boldsymbol{\theta}}(\rho, O)$), we will mostly focus on those of the form in equation (3), as the lessons learnt from this limited set can often be extrapolated to other, more general, scenarios. In either case, one usually uses a quantum device to estimate the loss and then leverages the power of classical optimizers to solve the optimization task:

$$\arg\min_{\boldsymbol{\theta}} \ell_{\boldsymbol{\theta}}(\rho, O). \quad (4)$$

Despite the relative simplicity of the variational quantum computing framework, solving equation (4) and training the parameters of a model can be a hard task. For example, the optimizer might get stuck in a suboptimal local minimum[12–18], a solution might not exist in the parameter hyperspace[19,20], or the landscape might exhibit BPs – the focus of this Review. Importantly, and as discussed subsequently, the hardness of training the loss will strongly depend on the inductive biases of the quantum algorithm (Box 1).

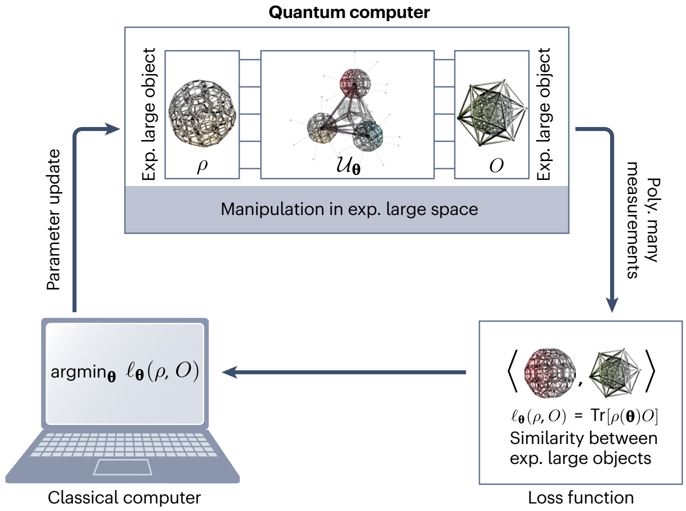

Fig. 1 | Variational quantum computing. An n-qubit quantum computer is initialized to a state ρ that is evolved through a parametrized quantum circuit $\mathcal{U}_{\boldsymbol{\theta}}$. At the output of the circuit, one performs a finite set of measurements used to estimate a loss function $\ell_{\boldsymbol{\theta}}(\rho, O) = \mathrm{Tr}[\mathcal{U}_{\boldsymbol{\theta}}(\rho)O]$. A classical computer takes as input the finite sample estimate of $\ell_{\boldsymbol{\theta}}(\rho, O)$ (or its partial derivatives) and attempts to find new sets of parameters that train the parametrized quantum circuit and minimize the loss. As such, by writing the loss in the vectorized form $\ell_{\boldsymbol{\theta}}(\rho, O) = \langle \rho(\boldsymbol{\theta}), O \rangle$, one can see that at the core of variational quantum computing lies the manipulation and comparison – using polynomially many samples – of the exponentially large operators $\rho(\boldsymbol{\theta}) = \mathcal{U}_{\boldsymbol{\theta}}(\rho)$ and O. Exp., exponentially; poly., polynomially.

Types of barren plateaus and loss concentration

We define a BP in a variational quantum computing scheme as occurring when the loss function, or its gradients, becomes exponentially concentrated around their mean as the number of qubits n increases. Intuitively, this means that the optimization landscape is mostly flat and featureless and that slightly changing the model's parameters $\boldsymbol{\theta}$ results in only an exponentially small change in $\ell_{\boldsymbol{\theta}}(\rho, O)$ or $\partial_\mu \ell_{\boldsymbol{\theta}}(\rho, O) = \partial \ell_{\boldsymbol{\theta}}(\rho, O)/\partial \theta_\mu$ (in which $\theta_\mu \in \boldsymbol{\theta}$). This creates challenges for minimizing the loss function, as optimizers determine descent directions by comparing the loss at different points. In particular, as one can only estimate expectation values from a quantum computer with a finite number of measurements N, the expectation values come with a statistical uncertainty that scales inversely proportional to $\sqrt{N}$ (Box 2). To resolve exponentially small changes and to optimize the loss, one would need an exponentially large number of measurements (shots), which makes the algorithm inefficient and non-scalable. Indeed, if a smaller number of shots is used, the optimizer will follow changes produced by the uninformative statistical fluctuations, which means the parameter updates will lead to a useless random walk in the loss landscape.

Probabilistic concentration and narrow gorges

The loss function concentration type most commonly found in the literature is probabilistic concentration. By defining $\mathbb{E}_{\boldsymbol{\theta}}$ as the expectation value with respect to $\boldsymbol{\theta}$ sampled from some domain according to some distribution, we say a loss exhibits a probabilistic BP if

$$\mathrm{Var}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)] \text{ or } \mathrm{Var}_{\boldsymbol{\theta}}[\partial_\mu \ell_{\boldsymbol{\theta}}(\rho, O)] \in \mathcal{O}\left(\frac{1}{b^n}\right), \quad (5)$$

for some $b > 1$, and for some (but not necessarily all) parameters $\theta_\mu \in \boldsymbol{\theta}$. As discussed subsequently, the choice of an ansatz $U(\boldsymbol{\theta})$, an initial state ρ and an observable O can individually induce the concentration. The exact value of the exponent base b implicitly depends on $U(\boldsymbol{\theta})$, ρ and O.

At this point, we note that one should understand probabilistic BPs as an average property of a loss landscape, which means that most points in the landscape should be exponentially flat. More formally, the exponential vanishing variance implies through Chebyshev's inequality that for any $\delta > 0$

$$\mathrm{Pr}_{\boldsymbol{\theta}}(|\ell_{\boldsymbol{\theta}}(\rho, O) - \mathbb{E}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)]| \geq \delta) \in \mathcal{O}\left(\frac{1}{b^n}\right), \quad (6)$$

or $\mathrm{Pr}(|\partial_\mu \ell_{\boldsymbol{\theta}}(\rho, O) - \mathbb{E}_{\boldsymbol{\theta}}[\partial_\mu \ell_{\boldsymbol{\theta}}(\rho, O)]| \geq \delta) \in \mathcal{O}\left(\frac{1}{b^n}\right)$. These equations show that the probability of the loss or its partial derivative deviating by more than δ from its mean is exponentially small, in which $\delta \in \Omega(1/\mathrm{poly}(n))$.

Here, we find it important to remark that although the definition of a BP in terms of the loss is clear, it becomes less clear when expressed in terms of partial derivatives. As mentioned, a BP arises when some, but not all, partial derivatives are concentrated, indicating that analysing the concentration of the overall loss is a more reliable way to diagnose a BP. In fact, it is entirely possible that a loss contains some partial derivatives that are concentrated, whereas others are not. Hence, care must be taken when establishing the absence of a BP by studying $\mathrm{Var}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)]$

Box 1 | Inductive biases

Inductive biases, broadly speaking, capture the assumptions and choices made when constructing the quantum algorithm. In the pipeline of variational quantum computing, these come in many forms. For example, when working with variational quantum algorithms, there are different degrees of freedom with regards to the choice of input state, but the measurement operator is usually more restricted. This is because in most variational quantum algorithms the problem information is encoded in the measurement operator (for example, the variational quantum eigensolver[117] determines the expectation value of the Hermitian operator for the desired ground state). In quantum machine learning, the input states tend to be fixed as the encoding scheme for classical data, the measurement operator and the choice of loss function are generally more flexible, offering additional design freedom. Similarly, although equation (1) shows a general parametrized quantum circuit, different instances of these computational models differ based on the inductive biases that influence $\mathcal{U}_{\boldsymbol{\theta}_l}$, that is, the choices in their specific form. This includes properties such as the number and types of gates, their placement and how the parameters are initialized.

as it is entirely possible that a non-concentrated loss arises only from a few well-behaved sets of parameters (and thus still has a BP). However, when every partial derivative is concentrated, one can prove that loss concentration and partial derivative concentration are equivalent[10,21]. Similarly, one can show that if first-order derivatives concentrate, then so do higher-order ones[22].

Although a probabilistic BP implies a largely featureless landscape, regions with large gradients around some minima (sometimes referred to as fertile valleys[23]) can still exist. The caveat is that these regions have to be small and the minima therein are always exponentially narrow (meaning that their relative volume in parameter space is always exponentially small[10]). These properties led researchers to name these minima narrow gorges. Although this term suggests a steep well, the slope of the walls in these gorges is bounded by the loss's Lipschitz constant, which is typically in $\mathcal{O}(\text{poly}\ (n))$ (ref. 10). Finally, as these are the most common BPs discussed in this Review, in what follows, we assume BPs to be probabilistic unless stated otherwise.

Deterministic concentration

Probabilistic BPs lead to mostly flat and featureless landscapes, but they still admit the presence of well-behaved regions such as narrow gorges. By contrast, there exists a second type of concentration, known as deterministic BPs, where all of the landscape is truly flat[24–27]. A deterministic BP arises when the deviation of the loss function from its mean is bounded for all parameter values, $\forall\, \boldsymbol{\theta}$, by an exponentially small term. That is, when

$$|\ell_{\boldsymbol{\theta}}(\rho, O) - \ell_0| \in \mathcal{O}\left(\frac{1}{b^n}\right), \tag{7}$$

for some $b > 1$, and where ℓ_0 is a concentration point that is usually independent of $\boldsymbol{\theta}$. Deterministic concentration thus arises because the state $\rho(\boldsymbol{\theta})$ only has exponentially small overlap with O for all values of $\boldsymbol{\theta}$, which implies that this BP phenomenon depends on the initial state and measurement operator. As further discussed subsequently, this situation can arise owing to high entanglement in an input state (see section 'Input states and measurements') and/or unital noise in PQC (see section 'Noise'). Clearly, equation (7) implies that for all parameter values $|\partial_\mu \ell_{\boldsymbol{\theta}}(\rho, O) - \ell_0| \in \mathcal{O}\left(\frac{1}{b^n}\right)$. A deterministically concentrated loss cannot have any extrema that are significantly separated from the loss's mean. In fact, equation (7) precludes any type of non-exponentially suppressed features in the landscape.

Methods for analysing variance

At the simplest level, one can determine whether the loss or its gradients are concentrating by evaluating randomly sampled parameters and computing how the variance scales with the number of qubits[8,16,28]. Similarly, one can compute other quantities that have been proposed as surrogates to study whether a BP occurs. These include the landscape's information content[29] and fluctuation[30], the local state's entanglement[31] or the landscape's Fourier coefficients[32–34]. Although these approaches can provide a rough estimate of scaling, they suffer from statistical errors owing to finite measurements of the loss and generally cannot distinguish between the two types of BPs.

A preferred, but more theoretically demanding, method for detecting BPs is to analytically study the scaling of the variance. In this method, assumptions about the PQC are necessary to make the problem more tractable. For example, if $\mathcal{U}_{\boldsymbol{\theta}}$ is a deep unitary circuit whose distribution of unitaries forms at least a 2-design[35] over a Lie group (for a brief introduction to the topic, see Box 3 for Lie algebras and Box 4 for an average over the group and for the PQC expressivity), one can use representation theory to exactly evaluate the variances[8,16,25,36–45]

Box 2 | Effects of finite sampling

In practice, one can only estimate the expectation value of the loss function $\ell_{\boldsymbol{\theta}}(\rho, O) = \mathbb{E}_{\rho(\boldsymbol{\theta})}[O]$ up to some finite precision determined by the number of measurement shots. To differentiate $\ell_{\boldsymbol{\theta}}(\rho, O)$ from its N-shot approximation, we use $\overline{\ell}_{\boldsymbol{\theta}}(\rho, O)$ to denote the latter. Assuming that the experiment repetitions used to obtain $\overline{\ell}_{\boldsymbol{\theta}}(\rho, O)$ are independent, one recovers the sample variance (denoted as Var_s, and also known as the standard deviation of the mean[303]) of the estimate:

$$\begin{aligned}\text{Var}_s[\overline{\ell}_{\boldsymbol{\theta}}(\rho, O)] &= \frac{\text{V ar}_{\rho(\boldsymbol{\theta})}[O]}{N} \\ &= \frac{\mathbb{E}_{\rho(\boldsymbol{\theta})}[O^2] - \mathbb{E}_{\rho(\boldsymbol{\theta})}[O]^2}{N}.\end{aligned} \tag{9}$$

Equation (9) quantifies the uncertainty to which $\overline{\ell}_{\boldsymbol{\theta}}(\rho, O)$ is computed. Clearly, there are two contributions to this uncertainty. The one in the numerator arises from the fact that $\rho(\boldsymbol{\theta})$ is generally not an eigenstate of O, and different measurements can therefore yield different outcomes. Second, the factor $1/N$ indicates that the expectation value is estimated with a finite number of samples and, using Chebyshev's inequality, this implies that the statistical fluctuations of the mean estimator are of order $1/\sqrt{N}$. Thus, in regions of a landscape with exponentially small gradients, exponentially many shots are typically required to determine a loss minimizing direction.

via Weingarten calculus[46,47]. Instead, for shallow circuits that do not form designs, one can instead assume that the local gates are sampled from a local group, thus mapping the variance evaluation to a Markov chain-like process which can be analysed analytically[48–60], numerically via Monte Carlo[61] and tensor networks[62,63], or studied via XZ calculus[64]. One can also take an iterative approach to explicitly integrate over the random parameters within a given angle range[23,56,65,66] or exploit tools from the Floquet theory[67,68]. In addition, if $\mathcal{U}_{\boldsymbol{\theta}}$ is a noisy channel, one can study the variance by making assumptions regarding to the type of noise acting throughout the PQC[24,69–72].

Origins of barren plateaus

Theoretical and numerical calculations can help to determine the flatness of the landscape, but they merely reveal symptoms of deeper issues in the variational quantum computing scheme. In this section, we discuss recent findings that argue that BPs appear as a form of curse of dimensionality, and we showcase how this phenomenon reveals itself under different scenarios. By understanding the connections between these different origins of BPs, we will see that one should be able to predict whether a model will exhibit an exponentially concentrated loss.

Curse of dimensionality

Recently, it has been argued that the very same feature that makes variational quantum computing appealing – the exponentially large dimension of the Hilbert space – is at the very core of the BP phenomenon[11,25,40,41,73,74]. To see that this is the case, let us begin by assuming that the PQC is a unitary as in equation (1). Then, a loss of the form in equation (3) can be expressed as $\ell_{\boldsymbol{\theta}}(\rho, O) = \langle \rho(\boldsymbol{\theta}), O\rangle = \langle \rho, O(\boldsymbol{\theta})\rangle$, where we introduced $O(\boldsymbol{\theta}) = U(\boldsymbol{\theta})^{\dagger} O U(\boldsymbol{\theta})$ and with $\langle A, B\rangle = \mathrm{Tr}[A^{\dagger} B]$. That is, the loss is the Hilbert–Schmidt inner product between two vectors in the exponentially large space $\mathcal{B}(\mathcal{H}_0)$. This simple realization shows that optimizing the loss can be understood as variationally trying to anti-align two exponentially large vectors via their inner product. This is already troublesome as – under quite general assumptions such as random initialization – the inner product between two exponentially large parametrized objects will be (on average) exponentially small and concentrated.

Now, of course, this does not mean that comparing any pair of exponentially large objects will always lead to exponentially small signals. Indeed, many classical machine-learning algorithms[75], as well as conventional quantum algorithms[76], manipulate vectors in the exponentially large Hilbert space by carefully choreographing the patterns of constructive and destructive interference of the probability amplitudes[77]. This is often because these methods incorporate specific structures or constraints that limit their expressiveness or control the exploration of the parameter space, thereby avoiding uncontrolled concentration effects. It is also worth noting that conventional quantum algorithms have even been adapted to establish advantage proofs for QML[78,79]. Yet, there are not many such algorithms in the variational setting as their construction requires carefully and purposely designed circuits with appropriate inductive biases for the task they are trying to solve. How well VQAs can be coerced into finding and exploiting such structure in the exponentially large Hilbert space is very much an open question.

In what follows, we will take a closer look at $\langle \rho(\boldsymbol{\theta}), O\rangle$ and illustrate how its different components can lead to a curse of dimensionality-induced exponential concentration owing to the choice of circuits, initial states, measurement operators and the effects of hardware noise.

Box 3 | From quantum circuits to Lie algebras and groups

The Lie subalgebras $\mathfrak{g} \subseteq \mathfrak{su}(\mathcal{H})$ have a fundamental role in barren plateau analysis. Recalling that $\mathfrak{su}(\mathcal{H})$ is the subspace of traceless anti-Hermitian linear operators on $\mathcal{H}$, any subspace $\mathfrak{g} \subseteq \mathfrak{su}(\mathcal{H})$ that is closed under commutation constitutes a subalgebra. Given a unitary noiseless parametrized quantum circuit as in equation (1) in which $U(\boldsymbol{\theta}_l) = e^{-i\theta_l H_l}$, its dynamical Lie algebra[16,118,304,305] is defined as

$$\mathfrak{g} = \langle \{iH_l\}_l \rangle_{\mathrm{Lie}} \subseteq \mathfrak{su}(\mathcal{H}), \tag{10}$$

in which $\langle S\rangle_{\mathrm{Lie}}$, for any subset $S \subseteq \mathfrak{su}(\mathcal{H})$, denotes its Lie closure, the real vector space spanned by every nested commutator between the elements of S. The dynamical Lie algebra is a fundamental object in the study of parametrized quantum circuits because its associated Lie group $G = e^{\mathfrak{g}}$ contains all unitaries that can ever be achieved by a choice of parameters $\boldsymbol{\theta}$ or number of layers L. One can explicitly verify this by using the Baker–Campbell–Hausdorff formula to re-express the composition of layers as the exponential of a linear combination of the nested commutators of the generators, which by definition belongs in $\mathfrak{g}$.

Circuit expressiveness

Perhaps, the main culprit responsible for the curse of dimensionality is the PQC as it has been shown that its expressive power (Box 4) is directly linked to BPs. Roughly speaking, the larger the space the PQC explores in an unbiased manner, the more it is prone – when randomly initialized – to lead to input states that have exponentially small inner products with the measurement operator. The previous connection between expressiveness and BPs has been mathematically formalized for unitary[8,16,25,40,41,48,80,81] and noisy circuits, as one can show that the more expressive a circuit is, the more the loss can concentrate. Similarly, the capacity of a PQC[82,83] – measured by how many functions it can represent – affects the concentration of the loss. The higher the capacity, the more likely it is that the loss will concentrate[84].

In particular, let us review the recent approach in refs. 25,40,41 for obtaining an exact analytical form for the loss function variance of deep unitary PQCs. By 'deep', we refer to the regime where the number of layers is large enough to guarantee that the distribution of unitaries corresponding to random parameter choices forms a two-design[35]. That is, that it approximately matches the Haar distribution on $e^{\mathfrak{g}}$ (in which $\mathfrak{g}$ is the circuit's dynamical Lie algebra, defined in Box 3) up to second moments with small additive approximation error. Assume for simplicity that either ρ or O belong to a group module $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H})$ that is non-trivial under the adjoint action of the circuit. It is worth recalling that a group module is a vector space closed under the action of the group. In the case of $\mathcal{B}$, $\mathcal{M}$ is a subspace that is closed under the adjoint action of $e^{\mathfrak{g}}$, consequently, if $A \in \mathcal{M}$, $UAU^{\dagger}$ also belongs to $\mathcal{M}$ for any $U \in e^{\mathfrak{g}}$ (Technically speaking, group modules are irreducible representations of the dynamical Lie group over the vector space $\mathcal{B}(\mathcal{H}_0)$.) One finds

$$\mathrm{Var}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)] = \frac{P_{\mathcal{M}}(\rho) P_{\mathcal{M}}(O)}{\dim(\mathcal{M})}. \tag{8}$$

Here, $P_{\mathcal{M}}(\rho)$ and $P_{\mathcal{M}}(O)$ are the norms of the projection of ρ and O onto the module $\mathcal{M}$. Explicitly, given a Hermitian orthonormal basis

Box 4 | Haar measure, Weingarten calculus and expressive power

Evaluating $\mathbb{E}_{\boldsymbol{\theta}}[\mathrm{Tr}[U(\boldsymbol{\theta})\rho U^{\dagger}(\boldsymbol{\theta})O]^t]$ can become tractable for unitary parametrized quantum circuits (PQCs) under certain assumptions. Since every set of parameters $\boldsymbol{\theta}$ determines a unitary through equation (1), one can evaluate the average over the induced set of unitaries $\mathcal{S}$, and its probability distribution dU. That is, one can evaluate $\mathbb{E}_{\boldsymbol{\theta}}[\mathrm{Tr}[U(\boldsymbol{\theta})\rho U^{\dagger}(\boldsymbol{\theta})O]^t]$ as

$$\int_{\mathcal{S}} \mathrm{d}U[\mathrm{Tr}[U\rho U^{\dagger}O]^t] = \mathrm{Tr}[\tau_{\mathcal{S}}^{(t)}(\rho^{\otimes t})O^{\otimes t}], \tag{11}$$

in which $\tau_{\mathcal{S}}^{(t)}(\cdot) = \int_{\mathcal{S}} \mathrm{d}U U^{\otimes t}(\cdot)(U^{*})^{\otimes t}$ is the tth fold twirl over $\mathcal{S}$. In the vectorized picture[46], one can also write equation (11) as $\langle\langle \rho^{\otimes t}|\hat{\tau}_{\mathcal{S}}^{(t)}|O^{\otimes t}\rangle\rangle$ with $\hat{\tau}_{\mathcal{S}}^{(t)} = \int_{\mathcal{S}} \mathrm{d}U U^{\otimes t} \otimes (U^{*})^{\otimes t}$ known as the tth moment operator. If $\mathcal{S}$ forms a group G, and if dU is the uniform Haar measure over this group, $\tau_{\mathcal{S}}^{(t)}$ (or $\hat{\tau}_{\mathcal{S}}^{(t)}$) can be evaluated analytically through the Weingarten calculus (see ref. 46 for a pedagogical introduction to this topic). In addition to capturing the statistical properties of the circuit, the operator $\tau_{\mathcal{S}}^{(t)}$ can be used to study the expressive power of a unitary PQC, that is, the breadth of unitaries that the circuit can produce. For example, one can quantify the potential, or ultimate expressiveness of the PQC by its dynamical Lie algebra $\mathfrak{g}$ defined in Box 3 as any unitary $U(\boldsymbol{\theta})$ produced by the PQC belongs in the dynamical Lie group $G = e^{\mathfrak{g}}$. Here, the larger the dimension of the algebra, the more expressive the circuit is, and controllable circuits are defined as those for which $\mathfrak{g} = \mathfrak{su}(2^n)$. Of course, shallow circuits will not be able to cover the whole dynamical Lie group G, as they can only produce unitaries in some subset $\mathcal{S} \subseteq G$. To deal with this situation, one can instead consider the current expressiveness which is quantified by the difference between $\tau_{\mathcal{S}}^{(t)}$ and $\tau_{G}^{(t)}$. If $|\tau_{\mathcal{S}}^{(2)} - \tau_{G}^{(2)}| \leq \varepsilon$ (in operator norm), then[35,40]

$$|\mathrm{Var}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)] - \mathrm{Var}_{G}[\ell_{\boldsymbol{\theta}}(\rho, O)]| \in \mathcal{O}(\varepsilon \, \|O\|_{\mathfrak{h}}) \tag{12}$$

shows that one can quantify how much the variance of a shallow circuit deviates from that of a Haar random PQC over G via ε.

for $\mathcal{M}$, denote $\{H_\mu\}_\mu$, then $P_{\mathcal{M}}(\rho) = \sum_\mu \mathrm{Tr}[H_\mu \rho]^2$. As one can see, and as further discussed subsequently, the three quantities on the right-hand side of equation (8) are associated with different scenarios in which BPs can arise. In particular, the expressive power of PQC is related to $\dim(\mathcal{M})$, the role of an input state and measurements are associated with $P_{\mathcal{M}}(\rho)$ and $P_{\mathcal{M}}(O)$, respectively. Having a PQC with exponentially large $\dim(\mathcal{M})$ and/or exponentially small $P_{\mathcal{M}}(\rho)$ and/or $P_{\mathcal{M}}(O)$ results in an exponentially small variance and therefore in BPs.

To finish, let us note that to obtain equation (8), one assumes that the circuit is deep enough so that it forms a two-design over $e^{\mathfrak{g}}$. The number of layers L for which this assumption is satisfied can be quantified with the help of the results in refs. 40,41. Interestingly, one can still bound how much the variance will deviate from that in equation (8) when the circuit is not a two-design (see Box 4 and the results in refs. 40,41,80,81). Moreover, for shallow circuits for which equation (8) does not hold, the computation of the loss function variance can become more intricate, but a similar idea holds: if the adjoint action of a circuit can move the measurement operator (or the initial state) in an exponentially large subspace of operator space, then a BP can arise.

Input states and measurements

Although highly expressive PQCs significantly contribute to BPs, circuits with limited expressiveness can also exhibit BPs depending on the initial state and measurement operator. For example, let us consider equation (8) again, and assume that the ensuing dynamical Lie algebra is not exponentially large, meaning that $\dim(\mathfrak{g}) \in \mathcal{O}(\mathrm{poly}\,(n))$. This occurs when $\mathcal{U}_{\boldsymbol{\theta}}$ has strong inductive biases such as encoding some symmetries (specific examples are discussed subsequently). Now, despite using a circuit with a non-exponentially large underlying algebra, the loss can still be exponentially concentrated. For example, if one picks O belonging to an exponentially large module (that is, small algebras can admit exponentially large modules), then one will find exponential concentration from the denominator in equation (8). However, if O belongs to a module whose dimension grows polynomially with n, such as $iO \in \mathfrak{g}$, the variance's denominator $\dim(\mathcal{M}) = \dim(\mathfrak{g})$ will only decrease polynomially with n, and therefore the circuit expressiveness cannot be responsible for a BP. In this case, the loss is obtained by comparing objects in a polynomially large subspace, potentially avoiding the curse of dimensionality. However, a deterministic BP is still possible if the norm of the projection of ρ into the module is exponentially small, that is, if $\mathcal{P}_{\mathcal{M}}(\rho) \in \mathcal{O}(1/2^n)$. Box 5 presents an example in which an inexpressive circuit can still lead to a BP, either because O belongs to an exponentially large $\mathcal{M}$ or because O comes from a polynomial module but the state is misaligned.

We refer the reader to refs. 25,40 for a conceptual discussion on the interplay among PQC, measurement and initial state. Therein, it was argued that the module purities $\mathcal{P}_{\mathcal{M}}(\rho)$ can be considered as generalized forms of entanglement[85,86], thus providing operational meaning to those quantities. Moreover, refs. 48,49,87–94 and refs. 31,37,38,95 studied problems in which the measurement operator or the state, respectively, can determine the presence of a BP. Then, ref. 96 showed that mid-circuit measurements can change the amount of entanglement in the state and thus determine whether a BP will arise. For the case of variational quantum computing with classical data, it has been shown that the encoding scheme can lead to initial states that are ill-aligned with the PQC and measurement operator, further highlighting the importance of choosing a well-design data embedding scheme[27,42,82,97–100].

Noise

In the previous sections, we considered unitary circuits, but the presence of hardware noise throughout the PQC can substantially affect the loss function optimization landscape and potentially induce BPs[24,69–72,101–103]. When modelling noise, the standard approach is to assume that the channels $\mathcal{U}_{\boldsymbol{\theta}_l}$ consist of a unitary parametrized gate followed by a noise channel $\mathcal{N}$ as $\mathcal{U}_{\boldsymbol{\theta}_l} = \mathcal{N}(U_l(\boldsymbol{\theta}_l)\rho U_l^{\dagger}(\boldsymbol{\theta}_l))$. In this picture, it has been shown that any noisy PQC whose noise channels have the maximally mixed state as its fixed point (such that $\mathcal{N}\circ\cdots\circ\mathcal{N}(\rho) \sim \mathbb{1}/2^n$) will induce a deterministic BP in the regime of a large number of layers[24,101]. Recent findings have shown that non-unital noise or intermediate measurements can alleviate the BP issue, suggesting that the relationship among noise, measurements and loss function concentration remains an open question[69,103–106].

Although it is not obvious a priori, one can see that in some cases noise-induced BPs still constitute a form of curse of dimensionality. For example, by recalling that noisy processes can always be understood as an entangling operation with some environment of n_E qubits. Thus, if one purifies[76,107] the noisy state $\rho_{\boldsymbol{\theta}} = \mathrm{Tr}_E[V_{\boldsymbol{\theta}}(|0\rangle\langle 0|^{\otimes n_E} \otimes \rho)V_{\boldsymbol{\theta}}^{\dagger}]$, one notices that the loss becomes $\ell_{\boldsymbol{\theta}}(\rho, O) = \langle V_{\boldsymbol{\theta}}(|0\rangle\langle 0|^{\otimes n_E} \otimes \rho)V_{\boldsymbol{\theta}}^{\dagger}, \mathbb{1}_E \otimes O\rangle$, with $\mathbb{1}_E$ the identity over the environment's Hilbert space. Thus, noise leads to an extended PQC $V_{\boldsymbol{\theta}}$ that can act in a larger space, which could be exponentially large (although not necessarily) and thus substantially aggravates the curse of dimensionality.

Architectures prone to barren plateaus

In the previous section, we have provided some intuition regarding the origin of BPs. In this section, we review some BP-prone architectures. We note that in most cases, deep versions of these architectures can lead to exponentially concentrated losses owing to their large dynamical Lie algebras. However, these architectures can still be useful in shallow circuits or when using smart initialization strategies.

Hardware-efficient ansatz and deep unstructured circuits

The generic term 'hardware-efficient ansatz'[108] is used for unstructured PQCs composed of single-qubit rotation gates interleaved with (fixed or parametrized) entangling gates. In this context, 'hardware-efficient' refers to the fact that the gates of the circuit are chosen not because they encode some strong inductive bias from the problem one is trying to solve, but because they are easy to implement on the device. For example, one only picks entangling gates that follow the device connectivity to avoid compiling overhead[87,109,110]. Under quite general conditions, a hardware-efficient ansatz will be universal[16], meaning that their dynamical Lie algebra is $\mathfrak{g} = \mathfrak{su}(2^n)$. As such, deep versions of these circuits are as expressive as possible and therefore will exhibit BPs irrespective of the initial state or measurement operator[8,36,48,111–114]. Here, it is important to note that several standard architectures exist in the literature, such as entangling gates acting on a brick-like fashion on alternating pairs of qubits or on a ladder. Although these can indeed exhibit different performances in their shallower forms, they will still lead to a BP if the number of layers becomes too large.

Examples of the problem-inspired ansatz

Just like the term hardware-efficient is used for a wide range of problem-agnostic circuits, 'problem-inspired ansatz' is a blanket term used for models that embed inductive biases about the problem at hand into the circuit architecture itself. In most cases, a problem-inspired ansatz adjusts circuit expressiveness by using specific sets of gates, distinct entangling gate topologies or by aligning parameters with the symmetries of the task. Here, one can generally expect that $\mathfrak{g}$ will be smaller than $\mathfrak{su}(2^n)$ (ref. 40). Although we will subsequently present circuits that can in fact sufficiently reduce the circuit expressive power to potentially avoid BPs, we here discuss problem-inspired architectures that can still have exponentially concentrated loss functions, despite their inductive biases.

Hamiltonian variational ansatz. Hamiltonian variational ansatzes[16,67,115,116] were originally introduced in the context of the variational quantum eigensolver[117]. Here, the goal is to find the ground state of a Hamiltonian H that we assume can be expressed as a sum of non-commuting terms $H = \sum_s H_s$ (that is, $[H_s, H_{s'}] \neq 0$ if $s \neq s'$). The Hamiltonian variational ansatz circuit is expressed as $U(\boldsymbol{\theta}) = \prod_l \prod_s e^{-i\theta_{ls}H_s}$, from where we can clearly see that the ensuing dynamical Lie algebra will strongly depend on what the Hamiltonian of interest H is. As such, the presence or absence of BPs will be extremely problem-dependent and a case-by-case analysis will be neccessary. For example, recent results have shown that the vast majority of one-dimensional translational invariant Hamiltonians H will lead to exponentially large algebras and thus to BPs[118].

Quantum alternating operator ansatz. The quantum alternating operator ansatz[119,120] is widely used to variationally attempt to solve combinatorial optimization problems. Here, one wishes to find the ground state of a Hamiltonian H_P that is diagonal in the computational basis and that encodes in its ground state the solution to some combinatorial problem of interest. The quantum alternating operator ansatz is constructed as $U(\boldsymbol{\theta}) = \prod_l e^{-i\theta_{lM}H_M}e^{-i\theta_{lP}H_P}$, in which H_M is the so-called mixer Hamiltonian[121–123] usually taken to be $H = \sum_{j=1}^{n} X_j$. The presence of BPs in a quantum alternating operator ansatz for maximum-cut tasks has been analysed in refs. 16,124,125, in which it has been shown that the loss will exhibit a BP for the vast majority of graphs if the circuit is deep enough. Nevertheless, the depth for which a quantum alternating operator ansatz can obtain useful solution and outperform state-of-the-art classical methods[20,126–128] remains an open question. Moreover, this exponential concentration phenomenon has been also studied in quantum alternating operator ansatz variants such as the multi-angle quantum alternating operator ansatz where every gate in the circuit is assigned its own parameter[129]. Here, one can show that for all graphs with non-trivial cut value, the circuit's dynamical Lie algebra dimension will be exponentially large implying that the loss will be concentrated[125,130–132]. Notably, although the depth at which BPs will occur depends on the graph, and will likely require a polynomial number[40,133], there are cases in which a quantum alternating operator ansatz can have BPs even with a depth of one[125]. With the previous being said, there have been promising research direction to train a quantum alternating operator ansatz in regimes where the BP proofs do not hold (for exaxmple, training classically on small problems and transferring parameters)[134–139]. However, the performance of these approaches at scale is still an open question.

Box 5 | Single-qubit rotation parametrized quantum circuits can have barren plateaus

Consider a unitary circuit composed of general single-qubit gates as $U(\boldsymbol{\theta}) = \prod_{j=1}^{n} U_j(\boldsymbol{\theta}_j)$, in which $U_j(\boldsymbol{\theta}_j)$ acts only on the jth qubit. This circuit is fairly inexpressive as it does not even generate entanglement. Moreover, its dynamical Lie algebra is simply given by $\mathfrak{g} = \oplus_{j=1}^{n} \mathfrak{su}(2)$ and thus of dimension $\dim(\mathfrak{g}) = 3n \in \mathcal{O}(\mathrm{poly}\,(n))$. Yet, if $O = Z^{\otimes n}$, then the operator belongs to an exponentially large module composed of all 3^n Pauli operators acting non-trivially on all the qubits. Assuming that each $U_j(\boldsymbol{\theta}_j)$ forms a two-design over $\mathbb{SU}(2)$ (ref. 40), a direct calculation reveals that for all ρ, $\mathrm{Var}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)] \in \mathcal{O}(1/2^n)$. However, if $O = Z_1$, then the measurement belongs to a module composed of the three Pauli operators acting on the first qubit. Now, equation (8) leads to $\mathrm{Var}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)] = 2(\mathrm{Tr}[\rho_1^2] - 1/2)/3$, with ρ_1 the reduced state of ρ on the first qubit. The choice of measurement operator can lead to a barren plateau (BP). When $O = Z^{\otimes n}$, the loss is obtained by comparing operators that can move in exponentially large modules, whereas for $O = Z_1$, the circuit can only move in the subspace of operators acting only on the first qubit. However, although the parametrized quantum circuit and measurements are not responsible for BPs, the initial state can lead to an exponentially concentrated loss. In particular, if the state is too entangled and $(\mathrm{Tr}[\rho_1^2] - 1/2) \in \mathcal{O}(1/2^n)$ (which occurs if ρ satisfies a volume law of entanglement[26]), a deterministic BP occurs.

Unitary coupled cluster singles and doubles ansatz. The unitary coupled cluster singles and doubles (UCCSD) ansatz is a problem-inspired PQC used to find the ground state of a fermionic molecular Hamiltonian using the variational quantum eigensolver[117,140,141]. Here, it has been shown that standard versions of unitary coupled clusters composed of controlled single excitation gates are particle-conserving universal unitaries[57]. Indeed, the results in ref. 57 imply that a deep randomly initialized UCCSD ansatz will exhibit a BP if the state of interest lies within a subspace with a number of excitations that scale with the problem size (see ref. 142 for a numerical verification of this fact). In the next section, we will discuss how some of these issues could be avoided, for example, by using shallow circuits and good initializations strategies.

Strategies to avoid or mitigate barren plateaus

Since the discovery of BPs, attempts have been made to mitigate their detrimental effects. A common feature shared between the tools discussed here is that they break the assumptions leading to the curse of dimensionality. We will review different techniques that transform the loss into the inner product between two objects that live in polynomially – rather than in exponentially – large spaces[11] (Box 6). Although these approaches allow one to avoid BPs, they also make variational models vulnerable to classical simulation (for more discussion, see section 'Beyond barren plateaus in variational quantum computing'). Then we will review methods that mitigate BPs even when operating in exponentially large spaces. Here, we particularly refer the reader to the alternative initialization section, as this seems to be one of the most promising ways to jump start the training in a region of large gradients, rather than in a random place of the optimization landscape.

Shallow circuits

Perhaps, one of the simplest ways to avoid BPs is to use shallow PQCs. Here, even if the loss is concentrated for sufficiently deep versions of the circuit (either because it is too expressive or owing to the effects of noise), as long as the number of gates is sufficiently small, one can expect that the effect of both probabilistic and deterministic BPs to be reduced for certain measurements. In addition, although this strategy does not help to circumvent the deterministic BP arising from the input state, the shallow-depth PQCs can avoid the (unital) noise-induced BP. Examples include logarithmic-depth hardware-efficient ansatzes with local measurements[26,48,58], or quantum convolutional neural networks[50,64], which are shallow by design, but also classically simulable[143]. Intuitively, these circuits avoid BPs – even if randomly initialized – by constraining the space that the measurement operator can explore. These insights have been used in the literature to show that when using the variational quantum eigensolver, along with a shallow circuit, it is more effective to use problem-encoding schemes that lead to more local measurement operators[49,144].

It is important to note that although shallow circuits can provably have non-exponentially vanishing gradients, this is only true for certain measurements. For example, a logarithmic-depth hardware-efficient ansatz can still exhibit a BP for a non-local measurement[48]. The question still remains as to whether a good solution exists within the set of parameters. By using a small number of gates, it is entirely possible that the set of parameters obtained from the optimization does not correspond to a good enough solution of the problem[84]. This phenomenon is known as a reachability deficit[20,126]. In general, there is no simple way of knowing whether a model will suffer from reachability deficits and a case-by-case analysis is necessary. In addition, shallow circuits can also have spurious local minima[84,145,146], making it even harder to find any good solution that might, or might not, exist in the landscape.

Box 6 | Avoiding barren plateaus by working in small spaces

Recently, it has been noted that several strategies to avoid barren plateaus do so by effectively encoding the relevant dynamics in the loss function to a polynomially large subspace of the operator space, thus avoiding the curse of dimensionality that causes barren plateaus. As discussed in the main text, one can think about the adjoint action of a unitary parametrized quantum circuit over the initial state $U(\boldsymbol{\theta})\rho U^{\dagger}(\boldsymbol{\theta})$, or the measurement operator $U^{\dagger}(\boldsymbol{\theta})OU(\boldsymbol{\theta})$ as only being able to exactly, or mostly, explore a polynomially large subspace $\mathcal{B}_\lambda$ of $\mathcal{B}(\mathcal{H}_0)$. That is, $\dim(\mathcal{B}_\lambda) \in \mathcal{O}(\mathrm{poly}\,(n))$ denoting A^λ as the projection of an operator onto $\mathcal{B}_\lambda$, the loss function becomes $\ell_{\boldsymbol{\theta}}(\rho, O) = \langle \rho^\lambda_{\boldsymbol{\theta}}, O^\lambda \rangle = \langle \rho^\lambda, O^\lambda_{\boldsymbol{\theta}} \rangle$. Here, it is extremely important to note that $\mathcal{B}_\lambda$ is not necessarily a module of the Lie group associated with the quantum circuit. As such, the loss function variance will not always be given by equation (8). Still, as one can only move in a small subspace, we can expect that varying the parameters allows the variance to decay polynomially as $\Omega(1/\dim(\mathcal{B}_\lambda))$. The input state and the measurement operators still need to be well aligned with $\mathcal{B}_\lambda$ because if either ρ or O has exponentially small component in the subspace, the loss will be deterministically concentrated.

Small dynamical Lie algebras

In the previous section, we presented problem-inspired ansatzes that are still too expressive to avoid exponential concentration. However, there are cases of PQCs whose dynamical Lie algebra dimensions grow only polynomially with the number of qubits. In these cases, depending critically on the choice of initial state and measurement operator, the adjoint action of a circuit may result in small modules. This allows a circuit to avoid probabilistic expressiveness-induced BPs, even in its deeper forms, as described by equation (8).

A circuit's expressiveness can be reduced by using circuits with fewer qubits[144,147,148] or encoding sufficient inductive biases defining the problem. In this context, it has recently been pointed out that one particularly important source of inductive biases is the set of symmetries that a problem satisfies. That is, if one knows that the task's solution must respect some symmetry, one should construct variational models that only explore the symmetry-respecting solution space[6,47,149–154]. For example, it has been shown that permutation-equivariant unitary PQCs – circuits that remain invariant under qubit permutations – have a dynamical Lie algebra whose dimension scales as $\mathcal{O}(n^3)$ (refs. 42,155). Similarly, $U(1)$-equivariant circuits have been shown to avoid BPs when the Hamming weight of the initial state does not scale with n (refs. 16,156,157). We note that ref. 158 studied PQCs in which all the gates commute, meaning that the size of the algebra is exactly equal to the number of distinct gate generators.

Another archetypal example of a PQC with small dynamical Lie algebra is that of parametrized matchgate circuits (also known as Ising-model Hamiltonian variational ansatzes[157,159–164]), where one can find that $\mathfrak{g} \simeq \mathfrak{so}(2n)$ and therefore $\dim(\mathfrak{g}) \in \mathcal{O}(n^2)$ (refs. 16,25,118,165) (Box 7). These circuits can have a reduced expressive power as they can only implement free-fermionic evolutions. Along a similar line, linear optics continuous variable PQCs acting on coherent light in n modes have an associated algebra $\mathfrak{g} \simeq \mathfrak{o}(2n)$ (refs. 166,167), thus also potentially leading to BP-free loss functions. Interestingly, these models have also been explored in classical machine learning, in which they are known as efficient unitary neural networks[168,169], and where it has already been noted that they do not exhibit vanishing gradients at large depths. However, despite being able to avoid BPs, working with the polynomial dynamical Lie algebra dimension could lead to the classical simulability of the VQAs (see section 'Beyond barren plateaus in variational quantum computing').

Although the previous examples illustrate that one can indeed find circuits with polynomial-sized dynamical Lie algebras, and thus showing that their action is reducible to some polynomial-sized space, such cases are quite rare. In fact, as mentioned previously, most circuits will exhibit exponentially large algebras[16,118,125], and the search for other PQC architectures with small algebras is ongoing.

Variable structure ansatzes

Variable structured PQCs attempt to mitigate probabilistic expressiveness and noise-induced deterministic BPs by leveraging classical machine-learning protocols to iteratively grow the quantum circuit by placing or removing gates that lower the loss function and preserve large gradients. As such, these approaches are able to use highly expressive gate sets by only exploring specific regions of the ansatz architecture hyperspace. To this end, several machine-learning-aided strategies have been applied, including evolutionary algorithms, short-depth compilation algorithms and automachine learning[170–182]. One of the most promising and well-studied variable ansatz strategies is the ADAPT-VQE (and its variants for other circuits such as the quantum alternating operator ansatz)[172,173,183–187]. For example, one can construct a circuit for quantum chemistry applications with gates taken from some pool (for example, from those arising in UCCSD), and this strategy has led to hardware-friendly circuits with few entangling gates that focus on regions with large gradients, often resulting in good solutions.

Alternative initialization strategies

One of the main limitations of studying probabilistic BPs by analysing the scaling of $\mathrm{Var}_{\boldsymbol{\theta}}[\ell_{\boldsymbol{\theta}}(\rho, O)]$ is that these quantities only describe how concentrated the loss is on average across the whole landscape. Such information is of course important as it allows one to understand what to expect if one randomly initializes the model's parameters. However, randomly initializing the parameters is never a good idea, and the initialization method can be critical in determining the ultimate performance of the model.

This observation has motivated several strategies to initialize a variational quantum computing scheme. These include restricted small angle initializations (where one still randomly initializes the parameters but in a more structured way, such as in a special region)[65,66,68,188–196], pre-training via classical[172,197–199] or quantum[200,201] methods, parameter transfer[138,139,202–208] and iterative learning strategies[23]. For alternative iterative variational approaches, analogous lower bounds on the sample complexity have been derived using the McLachlan principle[209,210].

Here we highlight that some of these initialization strategies have shown heuristic success and seem to be one of the most promising avenues to mitigate BPs. In fact, classical machine-learning models can exhibit vanishing gradient issues just like their quantum counterpart (see subsequently for further discussion), and smart initializations have been one of the main tools. Additionally, in chemistry problems, such as finding a ground state of a molecular Hamiltonian, the initialization is typically done using a problem structure – for example, initializing with a mean field approach. We expect that this warm-start approach will have a fundamental role in the future of variational quantum computing. However, these strategies have limitations that must be considered[211,212]. These include concerns such as the fact that one can initialize in a region that might have large gradients, but is not well connected to a global minimum, or that might be classically simulable[11]. Indeed, several workarounds have been proposed to prevent a smartly initialized circuit from running into problematic regions of the optimization landscape (such as monitoring the reduced state's entanglement[31], or using state-of-the-art optimizers[17,213,214]).

Box 7 | Its all relative (to the modules)

Until very recently, barren plateau (BP) study was performed on a case-by-case basis, analysing one architecture at a time. Although such fragmented patchwork analysis was pivotal to the recent development of a unified theory of BPs[40], it led to several misconceptions in the literature in which lessons learnt in one architecture are taken to be generically true and extrapolated to other parametrized quantum circuits. These include claim that 'global' measurements such as $O_G = X_1Z_2 \ldots X_{n-1}Z_n$ (when O acts non-trivially on all the qubits) lead to BPs, whereas local measurements such as $O_L = X_{n/2}$ (only one qubit is measured) prevent them. Although these claims are indeed true for certain architectures such as logarithmic-depth hardware-efficient ansatzes[26,38,48] (or to the single-qubit rotation ansatz of Box 5), they are not generically true. For example, a parametrized matchgate circuit can avoid BPs with O_G, but will lead to exponentially concentrated loss functions if one picks O_L (ref. 25). To understand why those two cases behave so differently, it is necessary to (again) study the module, or subspace, to which the adjoint action of the parametrized quantum circuit leads when action over O (see also Box 6). For the single-qubit rotation ansatz, O_G belongs to an exponentially large module, whereas O_L belongs to a polynomially small one. This statement is fully reversed in the matchgate case as O_G is in a small module, but O_L is in an exponentially large one.

Strategies that cannot avoid BPs

Although many promising strategies have been devised to prevent or alleviate BPs, some intuitive approaches fail because they do not address the underlying causes of BPs. Here we review two such strategies.

It has been claimed that changing the optimization method makes it possible to navigate a randomly initialized BP landscape. For example, one might assume that despite suppressed gradients, it is possible to find a loss-minimizing direction by using higher-order gradient information, gradient-free methods, quantum natural gradients[215,216] or other quantum Fisher information-based schemes. However, studies have shown that these approaches still require an exponential number of measurements to obtain a loss-minimizing direction[22,27,217].

Similarly, given the critical limitations that noise imposes on VQAs, it is essential to investigate whether error mitigation techniques[218–220] can sufficiently denoise the loss function to find a loss-minimizing direction. Unfortunately, research has shown that in the worst cases, mitigating noise-induced BPs requires an exponential number of resources[101,221–223]. This remains true even when the circuit has sub-logarithmic depth in certain worst-case scenarios[224]. Importantly, we note that it is not known whether such a worst-case scenario is representative of average cases.

Exponential concentration elsewhere

The study of BPs has both informed and been influenced by other areas of quantum computing and machine learning. Here we briefly review some of these results.

Quantum generative models

In quantum generative modelling, one aims to optimize a quantum state such that its probability distribution, which results from some set of measurements, matches a target distribution[56,225–235]. Quantum generative models are implicit because they only provide access to samples, unlike explicit models that provide direct access to the underlying distributions. Correspondingly, one can draw a distinction between explicit losses, which are formulated explicitly in terms of the model and target probabilities, and implicit losses, which compare samples from the model and the training distribution[230]. For quantum circuit Born machines, refs. 90,230 showed that the tension between using an explicit loss, such as the Kullback–Leibler divergence, with an implicit model, leads to a BP. Conversely, implicit losses, such as the maximum mean discrepancy[228,230] and some losses used for quantum generative adversarial networks[56,90,235], can be viewed as the expectation value of an observable and so can be analysed with the tools discussed previously.

BPs have also been studied in quantum Boltzmann machines[233], in which a parametrized thermal quantum state model is trained by minimizing a log-likelihood loss. In this case, the landscape is convex and so it is possible to derive convergence guarantees with only polynomial sample complexity[234]. Inherently quantum strategies for generative modelling also present an interesting avenue for finding quantum generative models that avoid BPs[230,236].

Kernel-based quantum algorithms

Exponential concentration also arises in kernel-based schemes[97,237–244], in which a quantum computer is used as an enhanced feature space from which inner products or similarity measures between the data states can be estimated (similar to the kernel trick)[245–248]. Unlike PQCs, one strength of kernel methods is that for a given choice of a kernel function and data set, one is guaranteed to find the optimal trained model that minimizes the empirical loss since the training landscape is convex. However, the kernel values can only ever be estimated statistically using a polynomial number of shots. If the kernel is exponentially concentrated, then the empirically estimated kernels contain, with high probability, no meaningful information about the input data. Training the model with these estimated kernel values then leads to a trivial data-insensitive model whose predictions are independent of the input data[238].

Quantum optimal control

In quantum optimal control, exponentially concentrated optimization landscapes can arise[145], but their connection to BPs was only explored after the BP literature gained traction. The intrinsic connection between these two areas of study was first reported in ref. 249 and later in ref. 250. Then, in ref. 16, it was noted that tools from quantum optimal control, such as the study of the PQC's dynamical Lie algebra, can be used to diagnose BPs. Nowadays, some tools developed in variational quantum computing have started to see use in optimal control schemes[250–254], and the extent of this cross-fertilization is yet to be seen.

Trainability of tensor networks

Several works have studied BPs in tensor networks. For instance, it has been shown that tensor-network-based machine-learning models exhibit BPs for global but not local losses[52], and isometric tensor network states can avoid BPs[54]. Similarly, BPs have been studied in quantum circuits inspired by matrix product states, tree tensor networks and the multiscale entanglement renormalization ansatz[53,255].

Learnability

Insights from BPs have allowed the formulation of theorems on the difficulty of variationally learning typical random states[48], typical random unitaries[39,55] and the output distributions of random circuits[230]. There is an interesting commonality between these results and statistical query learning[256,257] no-go theorems for learning typical random states[258], random unitaries[259–261], the expectation of an observable over random states[14] and the output distributions of random quantum circuits[262]. In all cases, it is worth noting that although BPs are fundamentally an algorithm-dependent phenomenon (but are, in some sense, independent of the learning problem), tools from statistical query learning can provide algorithm-independent bounds on the hardness of learning a specific function class. Nonetheless, mathematically, there are close links between the two families of results, with the difficulty in both cases boiling down to a curse of dimensionality. For a thorough investigation of the link between BPs and statistical query learning no-go theorems, see ref. 261.

Beyond barren plateaus in variational quantum computing

The study of BPs does not fully address the challenges in ensuring that variational quantum computing models can achieve a quantum advantage. Here we briefly review some of the other questions that need to be addressed.

Local minima and solution quality

It has been shown that quantum landscapes can be plagued with exponentially many suboptimal local minima, potentially making their optimization extremely hard[12,13,15,17,146,263–267]. Critically, the local minima issue affects both shallow circuits, for which no good solutions exist, and deep circuits, for which a solution might exist but reaching it remains computationally difficult. For example, when learning an unknown unitary with a shallow 1D parametrized circuit, there are exponentially many suboptimal local minima[265]. Indeed, we know that if the PQC is too shallow, then spurious local minima can appear in the landscape because it does not have enough parameters to control all available directions. Although these minima can be removed by overparametrizing the PQC by adding more parameters, this could require exponentially deep circuits[84,145,212,267,268]. This kind of realization leads to a powerful insight: a randomly initialized circuit with $\mathcal{O}(\mathrm{poly}\,(n))$ parameters that has a BP because the dynamical Lie algebra is exponentially large (and thus the PQC is not overparametrized[84]) and cannot be trained, even allowing for infinite measurements and removing the sampling noise because the optimizer is exponentially likely to get stuck in a local minimum[84,145,212,266,268]. Addressing the problem of local minima will be fundamental to guaranteeing that a model is useful, which will be particularly important when using alternative-initialization techniques.

Classical simulability of BP-free models

As we have discussed earlier, many existing techniques that avoid the BP curse of dimensionality do so by encoding the relevant dynamics in some subspace that grows polynomially in the number of qubits n. As an exponentially large space is no longer being used, one should be able to simulate the information processing ability of the PQC and thus classically estimate the loss function[11] (note that for non-classically simulable initial states or measurements the quantum computer may be needed to gather some data in an initial data collection phase). This is possible for all parameter settings for shallow hardware-efficient ansatzes[269,270] and problem-inspired ansatzes with polynomial dynamical Lie algebras[86,201,271–273]. It is also possible for any randomly chosen parameter setting with high probability for a wide class of unstructured parameterized quantum circuits[274,275], QCNNs[143], noisy circuits[104,276] and small-angle initialization strategies[277].

Not having to implement a PQC on the quantum device could lead to algorithms that are better suited for near-term quantum computers as the data acquisition phase could be less noisy than fully implementing the PQC[11,278,279]. Moreover, the connection between classical simulability and trainability is a current area of analysis[280]. Recent results have shown that classical algorithms, when equipped with measurements from a quantum computer, can be more powerful than their fully classical counterparts when solving certain contrived tasks[237,281–284]. Still, even if it remains to be determined whether such separation holds for practical applications, this is a promising direction of research.

Finally, it is worth highlighting that there are numerous fully classical machine-learning approaches that tackle the same applications as variational quantum computing. These include variational tensor network methods[285,286] and neural network states[287–290] for solving ground state problems, simulating the dynamics of quantum systems and state tomography. Although they are limited by the intrinsic challenges of classically simulating quantum systems, these approaches naturally sidestep the BP problem completely.

Link with the vanishing gradient problem

A common question when studying BP phenomena is the relationship to vanishing gradient problems in classical deep learning. Identifying a precise connection between these vanishing gradients has been challenging. This is partly because there are several different types of vanishing gradients in classical settings. Here, we review some of the potential similarities and differences that may lead to a multitude of perspectives.

Differences between the two phenomena

The most well-known variety of classical vanishing gradients is the vanishing or exploding norm with depth, caused by growing or shrinking weights owing to the activation functions[291]. This phenomenon differs from BPs in variational quantum computing, where BPs, which are not noise-induced, appear as the number of qubits increases, not with increasing circuit depth. That is, although a curse of dimensionality exists for the quantum case, classically the problem would be more accurately described as a 'curse of depth'. In addition, it is worth noting that in the quantum setting, the BP phenomenon can be closely related to ρ and O, as different initial states and measurements can lead to a model having – or not having – exponentially concentrated loss functions[25,48].

Classical solutions to vanishing gradients

Many solutions to the vanishing of classical gradients with depth have been developed, including batch normalization[292], layerwise normalization[293] and the use of Re-LU activation functions[294]. However, these techniques might still fail in some situations[295,296]. As these solutions target the problem of shrinking weights at large depths owing to the nonlinearity of the activation function, they do not seem be directly applicable to the quantum case.

Some deep classical neural networks still experience problems even with the solutions listed earlier. In models such as recurrent neural networks, these effects can be examined as the input length increases to infinity, and one can analyse contributions to the current prediction from the infinite past. Unrolling the network leads to a model with infinite depth, where the problems of vanishing gradients are further magnified[295]. To ensure a signal survives from the infinite past, one has to conceive of unitary neural networks that conserve signal at every step, similar to the unitary operations of quantum computers[169]. The general strategy of preserving information from past layers to improve gradient signals is applicable to networks beyond recurrent neural networks, and these solutions include skip connections, residual connections and general long short-term memory models[296]. Strategies derived from the idea of carrying information forward more robustly could be emulated using simultaneous preparations of quantum states, but direct copying is challenging because of the no-cloning theorem.

Finally, initialization strategies that restrict the angle range[297–299] and pre-training methods[300] have proven effective classically for avoiding the vanishing gradient problem. Analogous methods have been shown to lead to larger gradients in quantum contexts[65–68,196,198]. However, given that quantum landscapes have been shown to have numerous local minima in under-parameterized regimes[14], this strategy might be seen as less promising in quantum contexts.

The role of exponentially large spaces

In relation to concentration of measure, it is often asserted that the dimension of quantum spaces can be much larger than classical spaces. However, the full picture is more subtle, as classical models can also explore exponentially large spaces. For example, one can classically optimize a 2^n-dimensional probability distribution on n bits using probabilistic parametrized bit-flips. Thus, it seems that the crucial element is not the size of the space explored by quantum models per se but rather how the space is used. Indeed, it is the hope of quantum computing that novel, less computationally intense, paths through similar spaces of problems may be found. A key step forward in resolving the problem of BPs is therefore not only a powerful ansatz but also a parametrization that has an inherent bias towards problems of interest.

The cost of precision

In the classical case, the value of the loss and gradient can often be obtained to precision ε with resources that scale as $\mathcal{O}(\log(1/\varepsilon))$, so that even if a gradient is rapidly vanishing, one may compensate with additional precision. By contrast, for most quantum models based on expected values, stochastic sampling is often the only option, and it comes at a cost of $1/\varepsilon^2$ or $1/\varepsilon$ with clever use of amplitude amplification. As a result, any problems of decaying gradients are often exponentially magnified in the quantum case. However, there are classical models that make use of stochastic sampling to determine the value of a loss and gradient[301,302], so this difficulty is not limited to quantum models. An interesting strategy to sidestep this problem in quantum models could be to take inspiration from the classical side and move away from expected values towards most-likely readout values or another deterministic output that can be reliably determined to high precision without sampling.

Implications and outlook

The community's efforts have led to big advances in understanding the BP phenomenon. Its primary causes have been identified and in essence all stem from a curse of dimensionality. There has also been progress in identifying architectures that circumvent BPs. However, the ultimate impact of the BP phenomenon on the scalability of variational quantum models is yet to be determined.

The remaining questions largely boil down to the fact that the BP phenomenon is an average case notion. If a landscape has a BP, then the probability that any randomly chosen parameter setting has non-negligible loss gradients is exponentially small. Given that quantum loss functions are precision-limited, that is, estimated from a finite number of measurement shots, the probability of obtaining an informative signal at any randomly chosen parameter setting is exponentially small. Nonetheless, in exponentially narrow regions, a BP landscape can exhibit substantial gradients. The question is whether these regions are easy to find, and even if one finds them, whether they are useful and can allow meaningful training. Addressing this requires understanding the interplay between BPs, local minima, expressivity limitations and, crucially, classical simulability — all of which are open questions that we urge our community to explore. Indeed, we believe that the field of variational quantum computing could benefit from taking a holistic view of all these difficulties and the causes that underpin them.

At this point, the question of what comes next arises. Classical machine learning has been driven by a combination of rigorous theory and well-informed heuristic strategies, leading to successes beyond those predictable by theory alone. However, the quantum hardware needed to perform large-scale heuristic implementations of variational quantum models does not exist yet. Indeed, the original motivation for studying BPs was to understand limitations that could prevent scaling variational models to non-classically simulable regimes. In this regard, the field of BP has successfully carved out settings where a quantum advantage will be unlikely. Still, much work remains to be done. We envision that the BP study will continue to serve as a guide to determine whether, and where, a practical variational quantum advantage exists, and to hint towards the development of new variational methods that go beyond the standard procedure sketched in Fig. 1.

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